

Embedded Systems for Muon Tomography:

Time to Digital Converter Implementation to Improve Spatial Resolution of Muon Impacts

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Abstract:

This project investigates the use of a Time-to-Digital Converter (TDC) for high-precision time-of-flight measurements in scintillator-based muon detection, with the goal of improving spatial resolution in muon tomography. Initial work validated the TDC in measurement mode 2—with less precise microsecond scale signal timing measurement—by firing a start and stop signal to the TDC and confirming that the time reported by the TDC matched times measured on an oscilloscope. With success time measurements in mode 2, the project then transitioned toward implementing measurement mode 1 for nanosecond-scale measurements that are required to have around 1 foot of spatial resolution on a muon impact. The high-precision required the development of a new testing method using controlled signal delays through BNC cable propagation and the design of a custom high-speed printed circuit board to improve signal integrity. Although hardware integration issues prevented full validation of mode 1 within the project timeline, this work established the testing methodology, hardware architecture, and measurement strategy necessary for future implementation. Successful completion of this system would provide more precise localization of muon impacts, increasing the resolution and surveying capability of muon tomography.

Semester Reflection:

The second half of the semester was a more critical application of using the TDC timing technology that showed viability through our experimentation in the first half of the semester. Focus was shifted from validating the Time-to-Digital Converter (TDC) in measurement mode 2 toward building the hardware and testing infrastructure necessary to implement measurement mode 1. While mode 2 successfully demonstrated that the TDC could produce reliable timing measurements for longer intervals, transitioning to mode 1 introduced a new set of engineering

challenges. Because mode 1 is designed for extremely short time interval measurements on the nanosecond scale, the previous testing methods were no longer sufficient. This required developing a new strategy for generating precise delay signals and creating hardware capable of supporting stable high-speed measurements.

One of the first challenges was creating controlled signal delays that were shorter than could be generated directly by the ESP32. The chosen strategy was to use lengths of BNC cable to create predictable propagation delays. Since electrical signals travel through coaxial cable at a known fraction of the speed of light, varying cable length provides a controllable method for introducing nanosecond-scale timing offsets. This simulates the small differences in arrival time that will eventually correspond to scintillator panel timing differences in the final detector application.

It became clear that breadboard testing would make it difficult to reliably and easily test mode 1 due to electrical noise, signal degradation, reflections, and instability for accurate high-speed timing measurements. As a result, a significant portion of the second half of the semester was dedicated to designing and manufacturing a custom printed circuit board specifically for TDC testing in measurement mode 1. The printed circuit board was designed in KiCad and required creating custom footprints for both the ESP32 and the TDC breakout board so they could be mounted cleanly using header pins. A buffer/line driver was added to increase the current-driving capability of the ESP32 output signal so that it could properly trigger the TDC with a 3.3V signal after traveling through the BNC transmission line. Proper 50 ohm termination was added at the Stop 2 input to match the characteristic impedance of the BNC cables and reduce signal reflections that could otherwise falsely trigger the TDC or damage components. Schottky diode protection was also incorporated to protect sensitive components

from voltage spikes. Special attention was given to high-speed PCB layout practices to improve signal integrity. Large ground planes were added to both the top and bottom layers of the board to minimize ground path length and create a stable electrical reference plane for fast digital signals and via stitching was used extensively to connect the two ground planes, lowering inductance and reducing electromagnetic interference. After manufacturing, the board was assembled with resistors, capacitors, the signal buffer, Schottky diode protection, and BNC connectors.

The first assembled board was successfully powered, but initial testing revealed that the ESP32 repeatedly reset during boot, preventing code execution. The ESP would immediately boot normally if disconnected from the PCB, so there is clearly a hardware issue that does not allow regular booting. Troubleshooting using a multimeter did not reveal an obvious short circuit, suggesting that the issue may be related either to a subtle soldering fault or excessive current draw from the combined TDC and buffer circuitry. Unfortunately, due to the ending of the semester time did not permit finding a solution to the issues with the board.

Continuation of this project would begin with assembling a second PCB with better quality construction to attempt to fix the ESP reboot issues. Once the booting issue is solved, testing in mode 1 will begin to verify the timing strategy, hardware, the program used to measure the results, and the accuracy of measurements. With successful results from mode 1 testing it ideally would be put to the test signals from muon detectors to see how it performs beyond the testing environment. With the technology accurately timing the difference in time of flight between signals coming from both ends of the muon detector, it will need more work to adapt it into the current experimental set up, but it would serve as a way to increase spatial resolution of muon impacts and reduce error in the surveying ability of muon tomography.

Midterm Reflection:

The semester can be conveniently divided into two distinct phases defined by the measurement mode currently being used on the Time-to-Digital Converter (TDC). The first half of the semester focused on using measurement mode 2, which measures longer time intervals with less precision. This mode was useful for testing the viability of the selected TDC board, learning how the hardware functions, and enabling easier troubleshooting while developing the communication and control system. The second half of the semester will transition to measurement mode 1, which allows for much shorter and more precise measurements. The goal moving forward is to apply the lessons learned from measurement mode 2 and adapt them to implement the measurement strategy required for the final application.

The first phase of the project was particularly challenging but valuable because the project is both original and highly applied, meaning solutions had to be developed creatively and tested carefully. The specific application involves using a TDC to measure differences in time-of-flight from scintillator panel signals in order to determine the location of muon impacts. Because this is a fairly specialized application, there is limited documentation on how to use the exact components in this configuration. As a result, it was necessary to develop a deep understanding of the hardware and take a “bare-metal” approach to programming the device registers and managing communication between the TDC and the microcontroller.

A significant portion of the learning process involved carefully studying the TDC datasheet and analyzing signals directly with an oscilloscope. This process helped develop a much deeper understanding of how electronic systems operate at a fundamental level. Although the project uses MicroPython, which is a relatively high-level language, working directly with

the device registers required manipulating individual bits to configure the TDC, send commands, and interpret measurement results. This significantly strengthened low-level programming skills and improved my understanding of how software interacts with hardware.

Another major area of learning came from troubleshooting the system when measurements did not initially behave as expected. There were several points where the TDC failed to produce accurate measurements, which required developing a systematic approach to debugging. Solving these issues involved carefully isolating each stage of the measurement process and verifying it individually. This included creating controlled test cases, analyzing timing signals on the oscilloscope, inspecting raw output data from the registers, and checking each physical connection in the circuit. Working through these issues reinforced the importance of iterative testing and methodical debugging when working with hardware systems.

Through this process, measurement mode 2 was successfully implemented and verified to produce reliable timing measurements for longer time intervals. Comparing these results with oscilloscope measurements confirmed the precision of the TDC and provided confidence that the system is functioning correctly. With this foundation in place, the next stage of the project will involve configuring measurement mode 1 to measure much shorter time intervals on the nanosecond scale. This mode requires a different measurement strategy using a single start signal and both stop channels to measure the differences in the results. Implementing it will allow the system to move closer to the intended application of measuring precise timing differences between scintillator signals.

Activity Log

Date	Hours	Description	Next Steps
1/16/2026	3	Began research on TDCs. Selected 2 potential options for further research.	Choose the TDC board to use.
1/27/2026	3	Began working with ESP32. Ordered both boards.	Learn SPI, test by having ESP communicate to itself.
1/30/2026	3	Got SPI working. Assembled circuit for TDC on breadboard.	Test communication with TDC.
2/3/2026	2	Testing communication with TDC failed. Troubleshoot circuitry.	Look deeper into the TDC datasheet and figure out exactly how the cheap board is accessing the pins on the TDC.
2/9/2026	5	Successfully got SPI working between TCD and ESP32. Was able to read and write registers. Went through TDC documentation to fully understand the chip, wrote all the functions necessary to initialize TCD properly for our use case by writing directly to the registers.	Test timing between signals from generating start and stop signals with ESP32.
2/17/2026 - 2/19/2026	8	Rewrote some communication functions that were sending wrong information to the registers. Confirmed all configuration registers were set up correctly for testing in measurement mode 2 (long time scales). Tested measurement with sending signals from ESP32. Troubleshoot signals on the oscilloscope; was unable to get the interrupt to trigger when a second signal was received.	Resolve the interrupt issue. Get a measurement for delta t.
2/26/2026	3	Discovered DIS pin was not connected to VCC. This solved the interrupt problem. Was able to measure delta t, but there seems to be an offset or calibration error.	Fix offset for measurement to reliable and accurately measure delta t.
3/3/2026	5	Investigated timing lots of different delta t's between signals. Discovered that in this measurement mode it may have some offset or error when measuring smaller delta t's but is very precise when	Next steps are to set up registers for measurement mode 1 and try measuring nano-second delta t's. There is a new strategy for this mode which requires firing one trigger

		measuring larger times in the hundreds of micro seconds. Using the oscilloscope it is clear the wait times and signal fire times are quite inaccurate, but by measuring delta t on the oscilloscope with cursors verified the precision of the TDC. Measurement mode 2 works great.	before the signals come in and then sending the 2 signals to the 2 different stop channels and calculating the difference between the results stored in the two corresponding registers.
3/15/2026	4	Wrote midterm progress report. Created a table of TDC measurement times and oscilloscope measurement times to verify the accuracy of the measurements. They are accurate for any delta t below the time out. Took photos to document the process of the research.	Previous next steps still apply.
3/18/2026	3	Worked on writing the code for the configuration registers for the TDC to be in mode 1. Realized we need a new way to generate signals with a very small delay, shorter than the ESP can generate. Determined using length of BNC cable to generate small delays. Two options for signal are using a dedicated signal generator or the ESP.	Decide how to generate the signal and the delays to measure. Make sure to reduce and reflection in the signal when entering BNC cable with 50 Ohm resistance.
3/25/2026	5	Decided to focus on creating a PCB to make testing in mode one easier to allow BNC cables to transmit the signals. Began creating a simple schematic with KiCad to layout the wiring for the PCB. Concerned with how to get a sharp signal, with small rise time and enough current to be able to trigger the TDC with 3.3V, either an external function generator is needed or something to boost the current on the ESP's signal.	Keep working on schematic and turn it into a PCB design. Find a solution to a strong and sharp signal.
4/1/2026	10	Finished schematic. Made custom footprints for ESP and TDC with header pins to allow them to connect to PCB. Implemented a buffer to increase current of the ESP generated signal. Designed the PCB, lots of iterations to make sure everything was right.	Finishing touches and reviewing PCB before getting it manufactured.
4/8/2026	7	Added grounding top plate and bottom	Double check board and send create

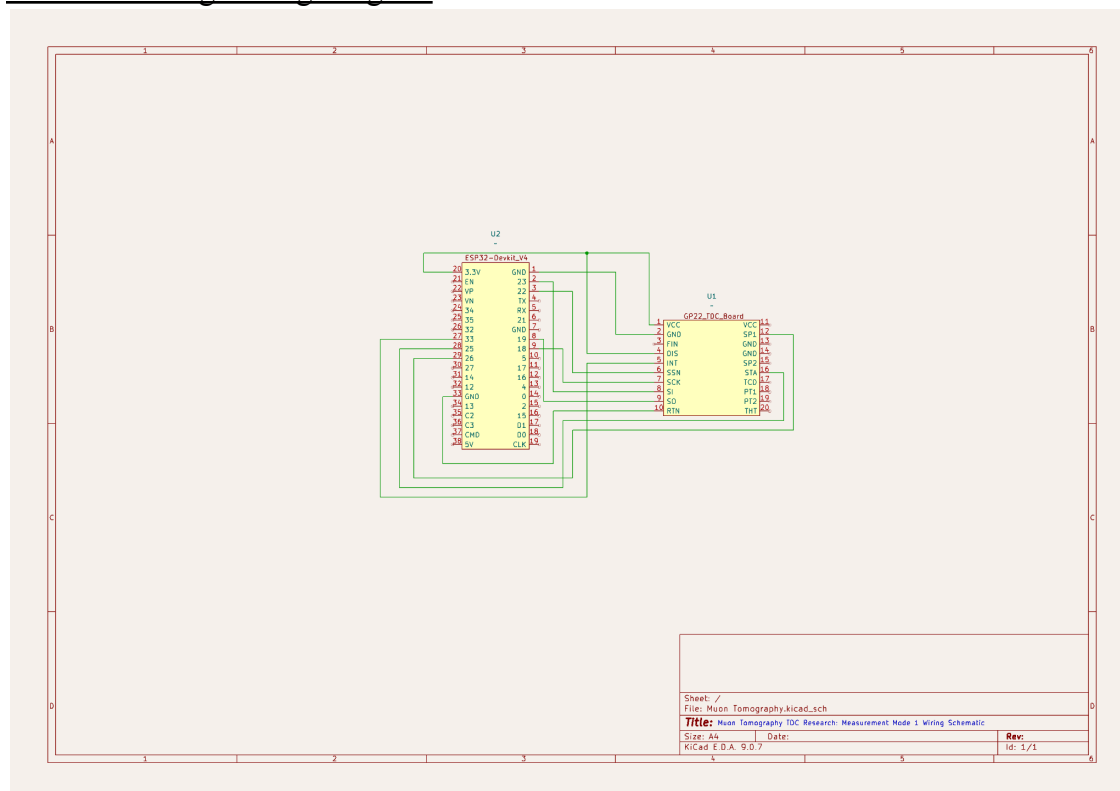
		plate to make ground paths as short as possible and make high speed signals more stable. Added via stitching to the board to make top and bottom ground as connected as possible and reduce inductance interference. Added protective Schotsky diode so components would not be harmed from any signal noise. Added 50 Ohm termination to the Stop 2 BNC connector to prevent reflections (50 Ohms to match the resistance of the cable). Added clearance rules and ran DRC tests to verify the board didn't have electrical errors.	Gerber files and inspect them. Then send it to the manufacturer.
4/15/2026	4	Created and checked gerber files for errors. Sent in PCB to be manufactured. The manufacturer noticed an error in the alignment of a few drill holes for vias on the board. Fixed the drill file issues and resent the gerber files to PCBWay to be manufactured and shipped.	Wait for PCBs to be manufactured and shipped. Order the rest of the components needed to assemble the board.
4/25/2026	2	PCBs arrived. Began connecting the resistors and capacitors that we had on hand in the lab.	Wait for the rest of the components to arrive and finish assembling the PCB.
4/29/2026	4	Received the last of the parts for the PCB including the buffer, Schotsky diode, and BNC connectors. BNC connectors with the correct footprint could not easily be found, so instead we drilled wider mounting holes and a hole to ground connector to bottom ground plate by scratching away the silk screen and soldering directly to the copper. With the holes drilled, the BNC connectors could be soldered to PCB.	Finish assembling PCB and place ESP32 and TDC into the headers. Power up ESP and check signals on an oscilloscope to see if everything is working properly.
4/30/2026	3	Finished assembling the first PCB. Soldered on remaining components. Powered on ESP32 and encountered error where ESP continuously resets and does not allow programs to be run. Attempted troubleshooting for a short with a multimeter and didn't find anything. May be a short or too much power being drawn from the ESP with the buffer and the TDC	Make a better second PCB with more careful soldering connections. See if the ESP will boot on the new PCB. Then test mode 1 with lengths of BNC cable to create signal delays.

		connected.	
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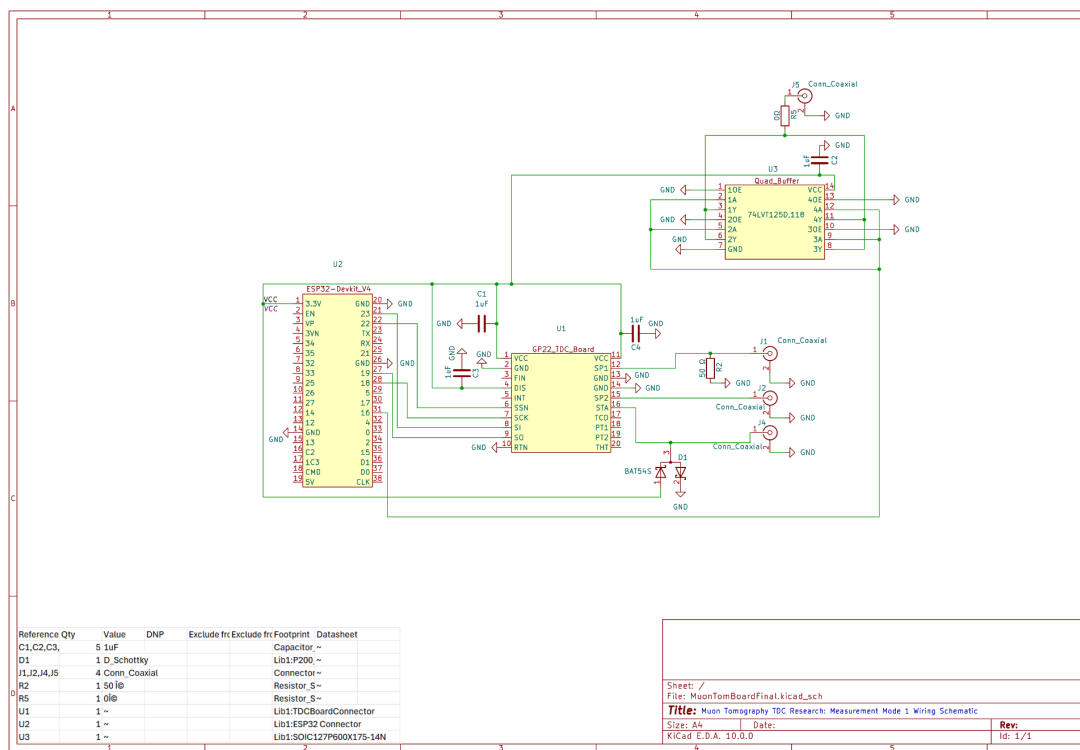
Mode 2 Measurements Testing:

Sleep Time (μs)	TDC Result ($\mu s \pm 0.25\mu s$)	Oscilloscope Measurement (μs)
1	69.5	69.6
20	84.75	84.8
50	114.5	115
100	164.5	165
500	569.25	569
1000	1068.75	1070

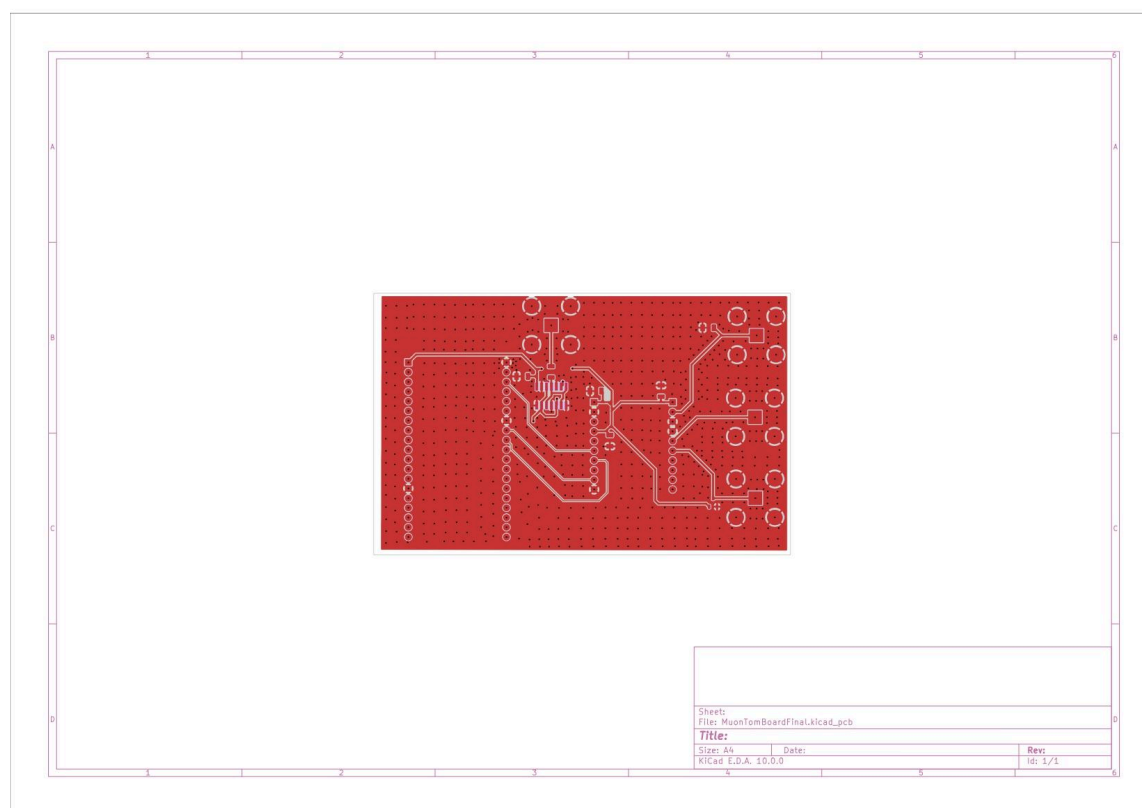
Mode 1: Testing Wiring Diagram



Mode 2: PCB Schematic



PCB Design Front



PCB Design Back

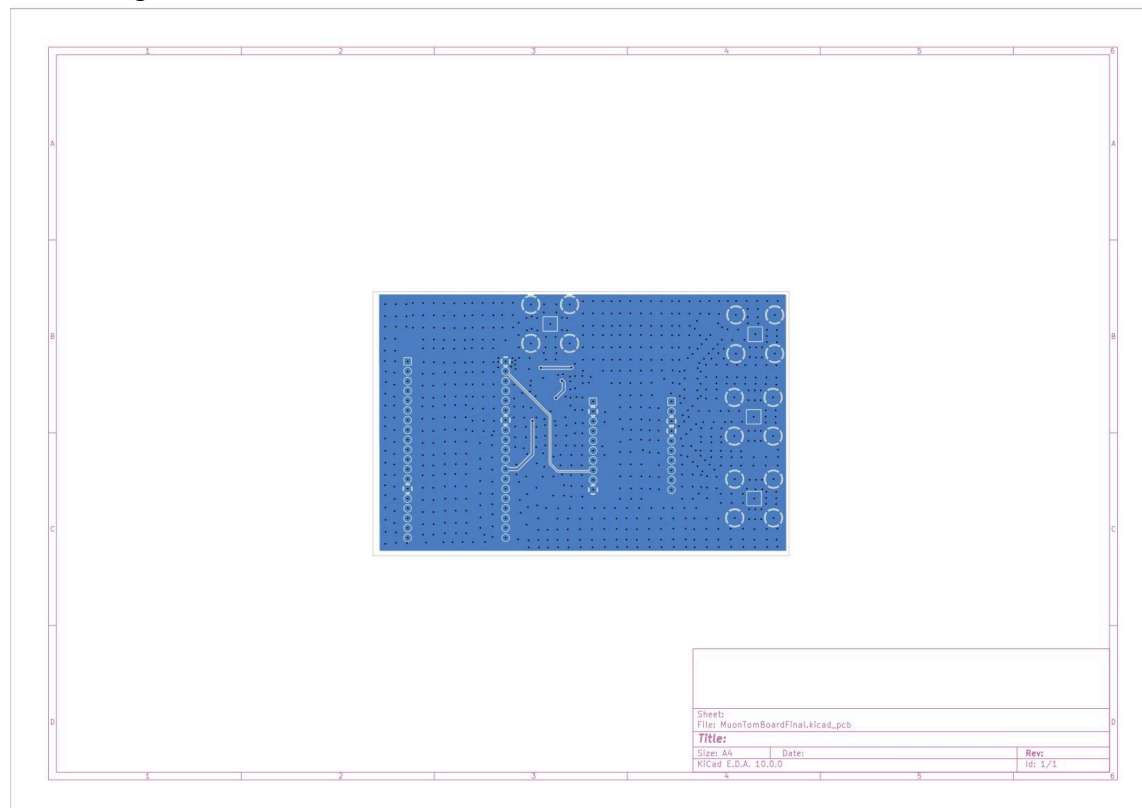
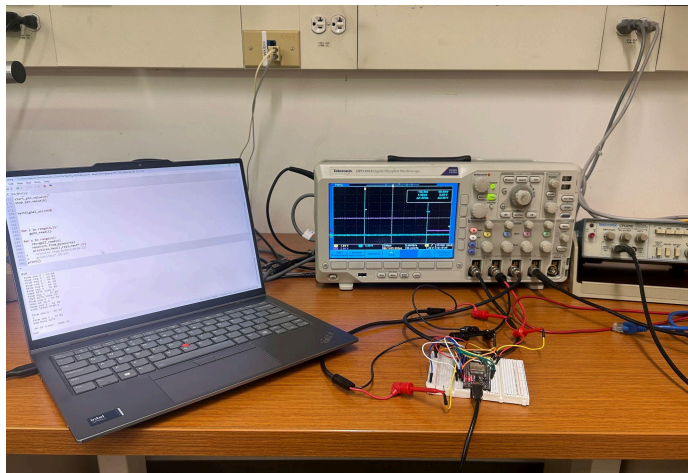
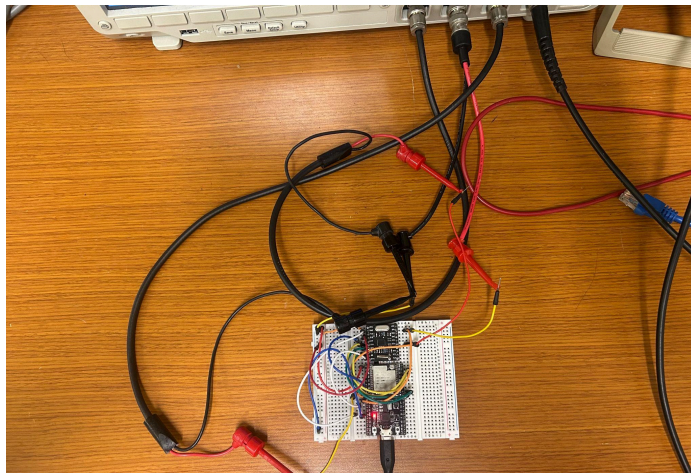


Photo Gallery:

Laboratory Testing Setup:



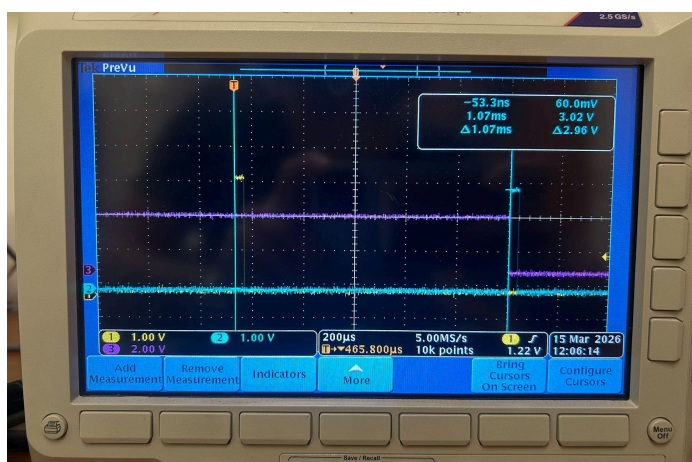
Board Connected to Oscilloscope for Testing:

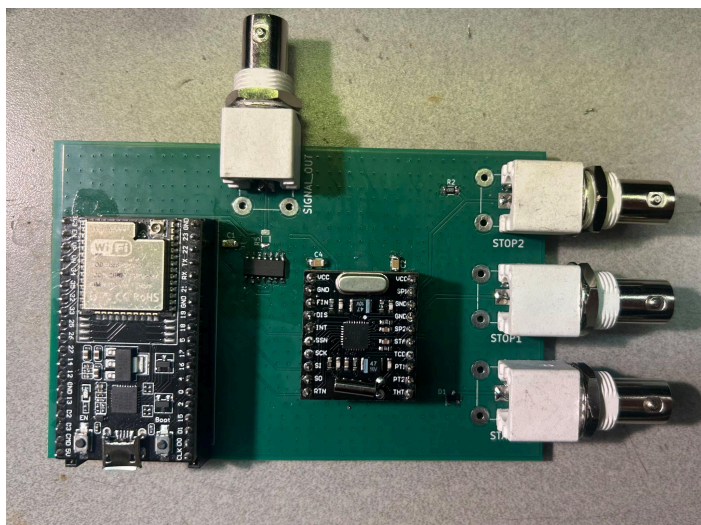
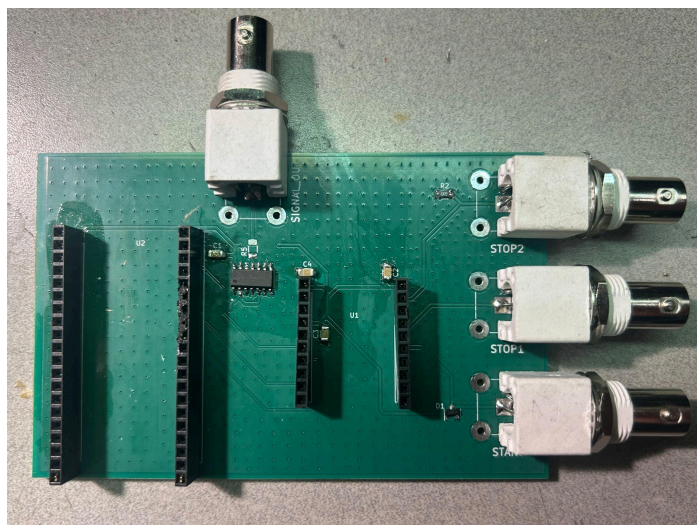
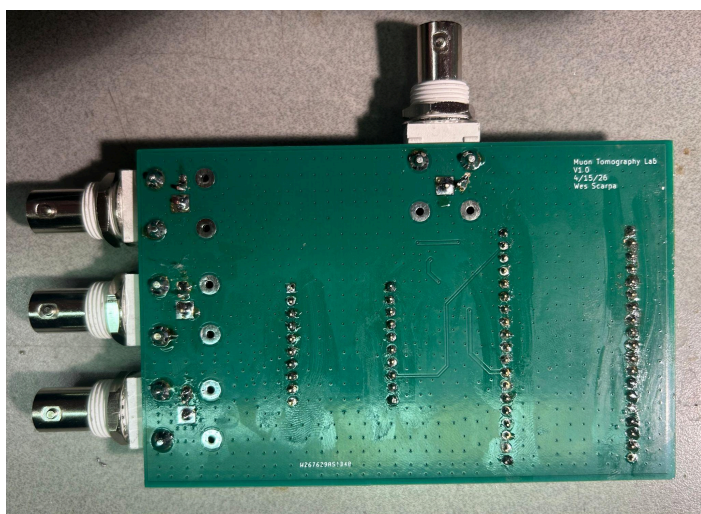


Oscilloscope Output for 164.5 μ s TDC Measurement:



Oscilloscope Output for 1068.75 μ s TDC Measurement:



Fully Assembled PCBPCB without ESP32 and TDCPCB Back

Bill of Materials

Component	Reference	Qty	Value	Footprint	Part Number
Capacitor	C1,C2,C3, C4,C5	5	1uF	Capacitor_SMD:C_0402_1206Metric	~
Protection Diodes	D1	1	~	Lib1:P200_SOT-23	BAT54S
Coaxial BNC Cable Connectors	J1,J2,J4,J5	4	~	Connector_Coaxial:BN C_TECconnectivity_1478035_Horizontal	~
Resistor	R2	1	50 Ω	Resistor_SMD:R_0402_1206Metric	~
Resistor (Place Holder)	R5	1	0 Ω	Resistor_SMD:R_0402_1206Metric	~
TDC Header	U1	1	~	Lib1:TDCBoardConnector	~
ESP32 Header	U2	1	~	Lib1:ESP32 Connector	~
Quad Buffer/Line Driver	U3	1	~	Lib1:SOIC127P600X175-14N	74LVT125D,118